Submission Summary

Conference Name
International Conference on Societal Intelligence and Smart Communities

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1583

Paper Title
DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES

Abstract
Conventional student attendance methods such as manual registers, physical sign-in sheets, fingerprint scanning, and QR-based systems are time-consuming, disruptive, vulnerable to proxy attendance, and difficult to manage for large classrooms. This paper presents the Multi-Facet Institutional Attendance System (MFIAS), a contactless, schedule-driven facial recognition system designed for automated and reliable attendance management. The system integrates an Automatic Departmental Timetable Engine to schedule classes and monitor attendance sessions without manual scanning. For identity verification, a multi-faceted computer vision pipeline handles practical variations such as dynamic classroom lighting, head pose variations up to ±15 degrees, eyeglasses, and beard-related appearance changes. The proposed pipeline uses Haar Cascade for face detection, the Dlib 68-point model for facial landmark alignment, RegNet-34 for facial feature extraction, and K-Nearest Neighbors (KNN, k=3) for identity classification. The system supports asynchronous batch verification of multiple students, accurate recording of arrival and departure times, duration calculation, and role-based authorization. Experimental evaluation using still images, live video, and previously recorded video demonstrates a recognition accuracy of 99.29%, a false omission rate of 0%, and an average verification time of 6 seconds. The system also generates tamper-resistant attendance logs and supports CSV, Excel, and PDF exports. The results demonstrate that MFIAS provides an efficient, contactless, and robust approach for institutional attendance automation.

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Submission Files

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